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Cross-Source Completion of Historical Chinese Incense Formulas: A Provenance-Blocked Benchmark with Source-Stable Co-Formulation Graphs

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Abstract

Chinese incense books transmit hundreds of aromatic formulas, but many survive only as partial ingredient lists, and each compilation has its own conventions. We ask a restoration question with a statistical core: given part of a formula, which missing materials, how many, and in what proportions can be inferred from other books? A provenance-indexed pipeline turns 592 source-linked candidates into 296 parsed compositions, 91 source-reviewed gold formulas from six work groups, and 146 genealogy-filtered weak records. Nested leave-one-work-out validation ensures that every test formula comes from a book unseen during training and tuning. We introduce source-stable conditional lift (SSCL), which pools within-work conditional log-odds and penalises their between-work spread, and show that it is the worst case of a transported edge over a dispersion-scaled uncertainty set. Conditioning on observed ingredients raised work-averaged hidden-ingredient F1 from 0.378 to 0.426, improving all five non-tied works (exact sign-flip $p = 0.0625$, the smallest value attainable; 90% interval 0.013–0.088). The dispersion penalty added no accuracy with six domains. Weak-record augmentation left count error unchanged at 1.145 and reduced SSCL F1 to 0.407; nested calibration cut the Brier score from 0.0427 to 0.0340. Materials seen fewer than five times in training are rarely recovered and limit accuracy.

Keywords: computational heritage; historical recipes; incense; set completion; compositional data; domain shift; weak supervision; calibration; randomisation inference

1. Introduction

Between the eleventh and the seventeenth centuries, Chinese scholars compiled incense books (*xiangpu*) that record how aromatic woods, resins, animal secretions and processed botanicals were combined into burnable formulas. The largest of these compilations copy, abridge and reorganise earlier books, so the same recipe can reach us in several witnesses, and many witnesses are incomplete: a title with a partial ingredient list, a list whose quantities are lost, or a later copy that silently drops a component. For heritage scientists who want to study, reconstruct or eventually reproduce historical scents, the first question is documentary rather than chemical. *Given the part of a formula that survives, what is most likely missing?*

Consider a formula for "seal incense" (*yinxiang*) attributed to the *Xiangpu* of Hong Chu. Its reviewed record lists eleven aromatic materials. Suppose a reader sees only six of them: agarwood, an aromatic grass, musk, onycha, patchouli and spikenard. Elsewhere in the tradition, Zhan-grade agarwood, borneol and white sandalwood habitually accompany these materials, so they are natural guesses for the missing part; the reviewed record

Received:
Revised:
Accepted:
Published:

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confirms all three, but it also contains frankincense and licorice, which such co-occurrence reasoning misses (Section 3.6). This paper turns that guess into a statistical prediction problem and asks how well it can be made when the evidence comes from *other* books. The distinction matters. Formulas inside one compilation share editorial conventions, vocabulary and a transmission history, so a model trained and tested on formulas from the same book can succeed by memorising the book rather than by learning transferable co-formulation structure.

Existing work does not answer this question. Historical smell research has built vocabularies, data models and archival frameworks for odours [1,2], and chemical analysis can reconstruct a scent when a physical residue survives [3]. Computational olfaction predicts perceptual descriptors from molecular structure, spectra or controlled mixtures [4–10], and inverse scent design optimises modern mixtures toward a target odour [11]. None of these settings involves incomplete historical records of natural materials whose composition varies with species, plant part and preparation. Conversely, item-based recommender models [12–14] and multi-label learning with missing labels [15,16] complete sets well but are usually evaluated with random splits that ignore where each record came from.

We therefore formulate formula restoration as *weighted set completion under source shift*. A model receives the observed materials of a formula together with their mass fractions; it must rank the hidden materials, predict the total number of materials, and distribute the remaining mass. Each historical work group is a source domain, and all evaluation is *work-blocked*: the test work is excluded from training, vocabulary construction, hyperparameter selection and probability calibration. Because only six work groups contain reviewed formulas, the work, not the formula, is the unit of statistical inference, and we analyse the consequences of this small number of clusters explicitly.

The paper makes four contributions.

1. **A provenance-blocked benchmark.** We release a hierarchy of evidence from 592 source-linked candidates to 91 source-reviewed formulas, with formula genealogy, work and collection identifiers, and deterministic masks for four observation mechanisms (Section 2.1).
2. **Source-stable conditional lift (SSCL).** SSCL estimates ingredient associations separately in each training work, converts them to marginal-adjusted log-odds lifts, pools them, and penalises their between-work spread. We show that the penalised edge is the worst case over an uncertainty set of transported lifts and, under a normal random-effects reading, a lower predictive quantile whose level is set by the penalty (Proposition 1).
3. **Provenance-aware weak supervision.** Following the weak-supervision principle of using noisy sources without granting them the status of expert labels [17], automatically parsed records enter training with a family-level weight budget and a collection block that prevents records from the test book from re-entering through a generic label (Section 2.5).
4. **Honest small-cluster inference and decomposed error analysis.** We report exact randomisation tests and confidence intervals over works, show that with six works and one tie this two-sided sign-flip test cannot reach $p < 0.0625$ (Proposition 3), give an identity that separates the guaranteed and the earned parts of the composition score (Lemma 1), and trace the remaining error to material rarity.

The main findings are as follows. Conditioning on observed ingredients transfers across books: every structured ranking model beats global frequency, and the directed co-formulation graph and SSCL raise work-averaged hidden-ingredient F1 by 0.048, improving every work in which the methods differ. The between-work penalty of SSCL

leaves accuracy unchanged, which we explain by the imprecision of dispersion estimates from at most five training works. Weak-record augmentation leaves size error unchanged and reduces SSCL ingredient recovery on the matched queries. Calibrated probabilities are markedly more reliable than raw graph scores. Most remaining error comes from materials that are rare or absent in the training works, which sets the recall ceiling for this closed-vocabulary task.

2. Materials and Methods

2.1. Sources, Curation and the Evidence Hierarchy

Source texts.

The corpus was assembled from public-domain Chinese incense books and their public transcriptions. Two notions of source are needed (Table 1). A *work group* is an attributed block of recipes: the formulas of Hong Chu's *Xiangpu*, Chen Jing's *Chen shi xiangpu*, Zhou Jiazhou's *Xiangcheng*, and three smaller recipe books (*Huizhai xiangpu*, *Liexiang xinpu* and *Moe xiaolu xiangpu*) that are transmitted inside the *Xiangcheng*. Work groups define the inferential unit because formulas within one group share terminology and transmission history. A *transcription collection* is the digital text from which records were extracted; three collections were used. The distinction matters for leakage: when the *Huizhai* group is tested, automatically extracted records from anywhere in the *Xiangcheng* transcription must also be withheld, because some of them may be unattributed copies of the test formulas.

Table 1. Work groups with source-reviewed (gold) formulas. "Evaluated" counts formulas with at least two materials in the vocabulary of the corresponding outer training fold.

Work group	Transcription collection	Gold formulas	Evaluated
Chen shi xiangpu (Chen)	*Chen shi xiangpu*	38	38
Xiangcheng (Xiangcheng)	*Xiangcheng*	29	29
Huizhai xiangpu (Huizhai)	*Xiangcheng*	10	9
Xiangpu of Hong Chu (Hong)	*Hong shi xiangpu*	6	6
Liexiang xinpu (Liexiang)	*Xiangcheng*	6	6
Moe xiaolu xiangpu (Xiaolu)	*Xiangcheng*	2	2
Total		91	90

Extraction.

Formula candidates were located in the three collections, yielding 592 source-linked candidates (332 in *Xiangcheng*, 244 in *Chen shi xiangpu*, 16 in *Hong shi xiangpu*). Each candidate keeps its work and collection identifiers, formula title, source span, a stable revision URL, material mentions, original quantity strings and processing notes. A deterministic parser assigned a mass to every aromatic material in 296 candidates, which we call *ratio-ready*. Traditional units were converted to a common relative unit; binders, liquids and processing aids were stored separately and excluded from the aromatic composition. Material mentions were mapped to 81 canonical material identifiers while keeping the source spelling; grade terms for agarwood (for example the Zhan-grade) remain distinct identifiers because they denote different commodities. Both steps face the problems familiar from historical named-entity recognition, such as spelling variants and shifting inventories [18], and from recipe extraction, where quantities must be attached to the right ingredient [19].

Genealogy.

Copying creates near-identical witnesses that would leak between training and test if treated as independent. Candidate pairs were retrieved by title-bigram overlap, material-

set Jaccard similarity, composition cosine similarity and textual similarity, and connected components of the resulting graph define provisional *genealogy families*; the 592 candidates form 415 families.

Source review.

A reviewer checked formula boundary, material identity and role, quantities, scope, genealogy and the normalised composition against the source text. Among 120 family-deduplicated candidates, 91 were accepted (75.8%); the 29 rejections involved quantity (13), material identity (10), scope (8), boundary (4) and role (4) problems, with some records failing several checks. The 91 accepted formulas, one per family, form the *gold* reference set. They contain 2 to 17 aromatic materials (mean 6.85, median 6; Figure 2b). The material distribution is long-tailed: 30 of the 81 materials occur in exactly one gold formula, and four occur in more than 40 (Figure 2c).

Weak records.

The 296 ratio-ready records decompose exactly into the 91 gold formulas, 59 further witnesses of gold families, and 146 records from 100 other families (Table 2, Figure 1a,b). The 59 witnesses are discarded, so that no copy of a gold formula can enter training through the weak channel. The remaining 146 records (84 from *Xiangcheng*, 58 from *Chen shi xiangpu*, 4 from *Hong shi xiangpu*) are *weak*: their parse succeeded but they were not reviewed. Together with the gold set they give 237 model-facing records in 191 families. Weak records are used only for training; every reported score is computed on gold formulas.

Table 2. Evidence layers. The ratio-ready layer is the disjoint union of the gold formulas, 59 witnesses belonging to gold families, and the weak records; the weak and gold layers share no family.

Layer	Records	Families	Role
Source-linked candidates	592	415	retrieval and coverage analysis
Ratio-ready records	296	191	parsed compositions; $296 = 91 + 59 + 146$
Gold (source-reviewed)	91	91	unit-weight training; all evaluation
Weak (eligible)	146	100	family-weighted training only
Model-facing union	237	191	gold $\cup$ weak

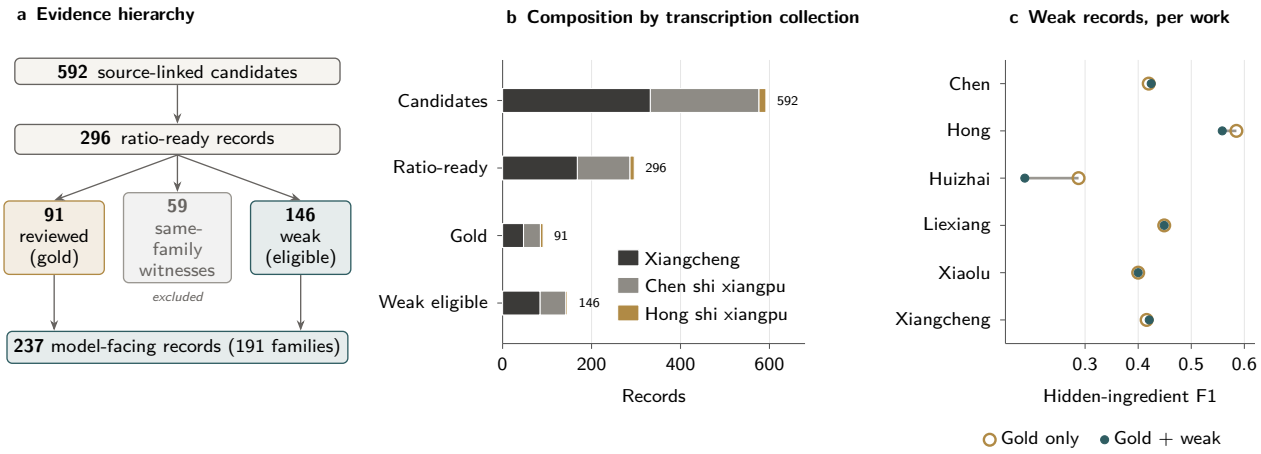


Figure 1. Evidence hierarchy and the weak-supervision test. (**a**) From 592 source-linked candidates to 237 model-facing records; witnesses of gold families are excluded before training. (**b**) Records per transcription collection at four stages; the gold bar counts formulas by the collection that transmits their work group. (**c**) Work-level hidden-ingredient F1 of SSCL trained on gold formulas only (open circles) or on gold plus weak records (filled circles), in the nested protocol of Section 2.11. Each point averages five masks within formula and then formulas within work. The work-averaged change is -0.0197 (exact sign-flip $p = 0.5$); masks are identical across training-evidence conditions.

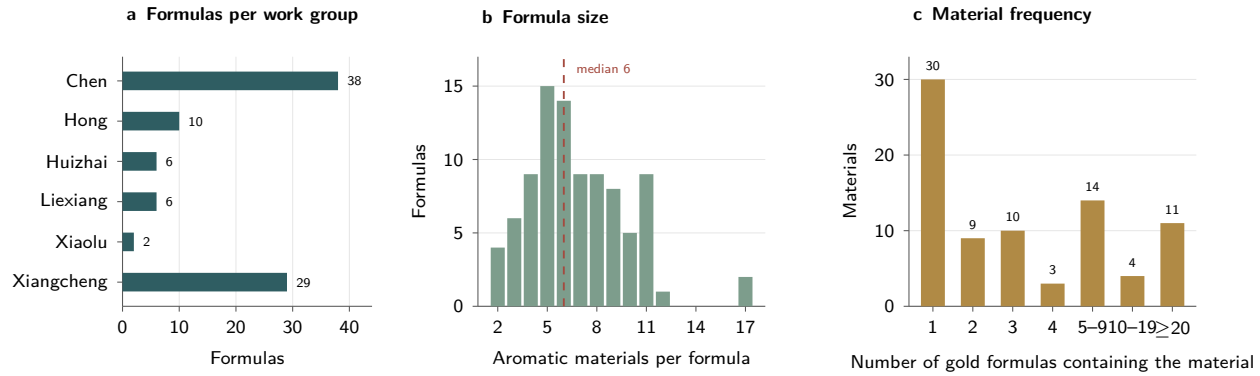


Figure 2. The 91-formula gold reference set. (**a**) Formulas per work group. (**b**) Number of aromatic materials per formula; the dashed line marks the median of six. (**c**) Number of materials by how many gold formulas contain them; 30 of 81 materials occur once.

2.2. The Completion Task

Formulas as compositions.

For formula i let $q_{ij} \geq 0$ be the converted aromatic mass of material $j \in \{1, \ldots, M\}$, $S_i = \{j : q_{ij} > 0\}$ its support and $K_i = |S_i|$ its size. The closed composition [20]

$$w_{ij} = \frac{q_{ij}}{\sum_{k \in S_i} q_{ik}}, \qquad \boldsymbol{w}_i \in \Delta^{M-1}, \tag{1}$$

discards the arbitrary total batch size, which varies across books, and keeps relative proportions.

Observation and target.

The model sees an observed subset $O_i \subset S_i$ together with the reference fractions $\{w_{ij} : j \in O_i\}$; the hidden set is $H_i = S_i \setminus O_i$. It returns (i) a score p_{ib} for each candidate $b \notin O_i$, (ii) a predicted size $\widehat{K}_i$ and hence a predicted hidden set $\widehat{H}_i$ of $\widehat{K}_i - |O_i|$ materials, and (iii)

a completed composition $\widehat{w}_i \in \Delta^{M-1}$ (Figure 3a). For a requested observed fraction f, the number of observed materials is

$$o(K, f) = \min\{K - 1,\ \max[1, \mathrm{round}(fK)]\}, \tag{2}$$

computed on the materials of formula i that belong to the training vocabulary of the fold, so at least one material is observed and at least one is hidden. Out-of-vocabulary materials always remain hidden.

Observation mechanisms.

Real damage is not uniform. Besides *uniform* retention, we simulate *abundance-biased* retention (probability of being observed proportional to reference mass, mimicking the survival of principal ingredients), *source-prefix* retention (the first o materials in source order, mimicking truncated copies), and uniform retention in which the count decoder is *not told* f. In the terminology of missing-data theory [21], the first and last mechanisms are completely at random with respect to material identity, whereas the abundance and prefix mechanisms are informative. The primary task uses uniform retention at $f = 0.5$; secondary analyses use $f \in \{0.25, 0.5, 0.75\}$. Masks are generated from hashed seeds, and five masks per formula are used throughout.

Estimand.

Let $\overline{F}_{md}$ be the hidden-ingredient F1 of model m (Section 2.12) averaged first over the five masks of each formula and then over the evaluated formulas of work d. The target of inference is the work-averaged difference from the frequency baseline,

$$\widehat{\Delta}_m = \frac{1}{D} \sum_{d=1}^{D} \Delta_{md}, \qquad \Delta_{md} = \overline{F}_{md} - \overline{F}_{\mathrm{freq},d}, \qquad D = 6. \tag{3}$$

Equal weights answer the question "how much does the method help for a new book?" and prevent the two large groups (Chen and Xiangcheng, 67 of 90 evaluated formulas) from determining the result. The formula-weighted average is reported as a sensitivity analysis.

2.3. Baselines: Frequency and the Directed Co-Formulation Graph

Let $X \in \{0, 1\}^{N \times M}$ be the presence matrix of the N training formulas and $c_b = \sum_i X_{ib}$. The *frequency* baseline ranks candidates by the smoothed prevalence

$$\pi_b = \frac{c_b + \frac{1}{2}}{N + 1}. \tag{4}$$

The *directed graph* conditions on the observed materials. With $c_{ab} = \sum_i X_{ia} X_{ib}$, its edge is the smoothed conditional frequency

$$G_{ab} = \frac{c_{ab} + \alpha \pi_b}{c_a + \alpha}, \tag{5}$$

an estimate of $\Pr(b \in S \mid a \in S)$ shrunk toward π_b with prior weight α. It is asymmetric ($G_{ab} \neq G_{ba}$ in general): a common base material predicts little about a rare modifier, whereas the modifier may predict the base strongly. For any edge matrix E, the score of candidate b is

$$p_{ib} = R\big(\{E_{ab} : a \in O_i\}\big), \qquad R \in \{\text{mean, max, mean of the top two}\}, \tag{6}$$

where the reducer R is a tuning parameter.

2.4. Source-Stable Conditional Lift

The directed graph pools all training formulas as if they came from one book. If a pairing is common in one compilation because of a single editor's habit, the pooled edge will over-predict it in a new book. SSCL instead estimates every association inside each training work, puts the works on a common scale, and asks how consistently the association holds (Figure 3c).

Step 1: within-work conditionals.

For training work d with N_d formulas, let c_{bd}, c_{ad} and c_{abd} be the corresponding counts. Work prevalence is shrunk toward the global prevalence,

$$\pi_{bd} = \frac{c_{bd} + \kappa \pi_b}{N_d + \kappa}, \tag{7}$$

and the within-work conditional probability $\theta_{abd} = \Pr(b \in S \mid a \in S, \text{ work } d)$ receives a beta prior centred on π_{bd}. Its posterior mean is

$$\theta_{abd} \mid \text{data} \sim \text{Beta}\big(c_{abd} + \alpha\pi_{bd},\ c_{ad} - c_{abd} + \alpha(1 - \pi_{bd})\big), \qquad \bar{\theta}_{abd} = \frac{c_{abd} + \alpha\pi_{bd}}{c_{ad} + \alpha}. \tag{8}$$

Step 2: marginal-adjusted lift.

Books differ in how often they use each material. To compare associations across books we remove the work-specific base rate on the log-odds scale:

$$\ell_{abd} = \operatorname{logit} \bar{\theta}_{abd} - \operatorname{logit} \pi_{bd}. \tag{9}$$

The lift is zero when a carries no information about b in work d. In particular, if a never occurs in work d ($c_{ad} = 0$) then $\bar{\theta}_{abd} = \pi_{bd}$ and $\ell_{abd} = 0$.

Step 3: pooling and dispersion.

Works in which a is rarely observed carry little information about its associations. With exposure weights $v_{ad} = c_{ad}/(c_{ad} + \alpha) \in [0, 1)$, which vanish for works where a is absent,

$$\mu_{ab} = \frac{\sum_d v_{ad} \ell_{abd}}{\sum_d v_{ad}}, \qquad s^2_{ab} = \frac{\sum_d v_{ad} \big(\ell_{abd} - \mu_{ab}\big)^2}{\sum_d v_{ad}}. \tag{10}$$

Step 4: transport and penalise.

The prevalence of b in the target book is unknown, so the pooled lift is added to the global log-odds, and the spread is subtracted:

$$G^\star_{ab} = \operatorname{expit}\big(\operatorname{logit} \pi_b + \mu_{ab} - \gamma s_{ab}\big), \qquad G^{\text{SS}}_{ab} = (1 - \lambda)\, G_{ab} + \lambda\, G^\star_{ab}, \tag{11}$$

with $\operatorname{expit}(x) = 1/(1 + e^{-x})$. The blend weight λ keeps the direct graph as a fallback for materials with little within-work support. Candidates are scored by Equation (6) with $E = G^{\text{SS}}$. The shrinkage constant is fixed at $\kappa = 5$; α, γ, λ and the reducer are selected by inner validation.

The following statement makes the role of the penalty precise.

Proposition 1 (Robust-edge representation). *Let* $\mathcal{U}_{ab}(\gamma) = \{\ell : |\ell - \mu_{ab}| \le \gamma s_{ab}\}$. *Then*

$$G^\star_{ab} = \min_{\ell \in \mathcal{U}_{ab}(\gamma)} \operatorname{expit}\big(\operatorname{logit} \pi_b + \ell\big). \tag{12}$$

For $s_{ab} > 0$, if the lift of the unseen target work satisfies $\ell_{ab,d^\star} \sim \mathcal{N}(\mu_{ab}, s_{ab}^2)$, then

$$\Pr\{\operatorname{expit}(\operatorname{logit}\pi_b + \ell_{ab,d^\star}) \geq G^\star_{ab}\} = \Phi(\gamma),$$

where Φ is the standard normal distribution function.

Proof. expit is strictly increasing, so the minimum over the interval is attained at its lower endpoint $\mu_{ab} - \gamma s_{ab}$. Under the normal model, $\Pr(\ell_{ab,d^\star} \geq \mu_{ab} - \gamma s_{ab}) = \Phi(\gamma)$, and monotonicity transfers the event to the edge scale. □

Thus $G^\star$ is an interval-robust edge [22]: it uses the least favourable lift in the specified interval. Under the normal random-effects assumption, the grid $\gamma \in \{0, 0.25, 0.5, 1\}$ gives lower predictive quantiles at approximately 0.50, 0.40, 0.31 and 0.16, respectively. If $s_{ab} = 0$, the interval reduces to a single point and the displayed probability is one. Setting $\gamma = 0$ gives a pooled lift without penalty, $\lambda = 0$ recovers the direct graph, and a material observed in only one training work has $s_{ab} = 0$.

Remark 1. *In a random-effects model $\ell_{abd} = \ell_{ab} + u_{abd} + e_{abd}$ with between-work effect u_{abd} (variance τ^2_{ab}) and estimation error e_{abd} (variance ς^2_{abd}), the dispersion s^2_{ab} in Equation (10) estimates approximately τ^2_{ab} plus a weighted average of the ς^2_{abd}. The penalty therefore combines genuine heterogeneity with sampling noise, which makes it conservative for poorly supported edges. A moment estimator of the DerSimonian–Laird type [23] would separate the two components. Moreover, with at most five training works the relative standard error of a standard deviation is roughly $1/\sqrt{2(D-1)} \approx 0.35$ under normality, so the penalty re-ranks candidates using a noisy quantity. Section 4.2 returns to this point.*

2.5. Provenance-Aware Weak Supervision

Let $z_i \in \{\text{gold}, \text{weak}\}$ be the evidence class of record i, $\varphi(i)$ its genealogy family, $C(i)$ its transcription collection, and $|\varphi|$ the number of weak records in family φ across all collections. Records receive the weights

$$\omega_i(\eta) = \begin{cases} 1, & z_i = \text{gold}, \\ \eta/|\varphi(i)|, & z_i = \text{weak}, \end{cases} \qquad \eta \in \{0, 0.1, 0.25, 0.5, 1\}, \tag{13}$$

and every count in Sections 2.3–2.4 is replaced by its weighted version, for example $c^{(\omega)}_b = \sum_i \omega_i X_{ib}$, $c^{(\omega)}_{ab} = \sum_i \omega_i X_{ia} X_{ib}$ and $N^{(\omega)} = \sum_i \omega_i$. The size histogram, the EASE Gram matrix and the composition neighbours use the same weights. Each weak collection enters SSCL as an additional source domain.

For an outer test work $d^\star$ with families $\mathcal{F}_{d^\star}$, the admissible weak pool is

$$\mathcal{W}_{d^\star} = \{i : z_i = \text{weak},\ C(i) \neq C(d^\star),\ \varphi(i) \notin \mathcal{F}_{d^\star}\}. \tag{14}$$

The same rule is applied to every inner validation work. The weight η is selected jointly with the model hyper-parameters by inner validation.

Proposition 2 (Family budget). *For every family φ, the total weight of its admissible members is $\sum_{i \in \varphi \cap \mathcal{W}_{d^\star}} \omega_i = \eta\,|\varphi \cap \mathcal{W}_{d^\star}|/|\varphi| \leq \eta$. Consequently the effective weak mass*

$$N^{\text{eff}}_{\text{weak}}(d^\star) = \sum_{i \in \mathcal{W}_{d^\star}} \omega_i \tag{15}$$

is at most η times the number of families represented in $\mathcal{W}_{d^\star}$, and $\eta = 0$ reproduces gold-only training exactly.

Proof. Sum Equation (13) over the admissible members of φ; the ratio of admissible to total members is at most one. For $\eta = 0$ all weak weights vanish and every weighted count equals its gold-only value. □

With $\eta \leq 1$, a family of copied witnesses can therefore never weigh more than one reviewed formula, and a family whose members are partly blocked by the collection rule contributes proportionally less.

2.6. Item-Based Comparators

Three standard recommender models test whether other ways of exploiting item dependencies do better. EASE [12] learns a dense item-to-item matrix in closed form: with $W = \mathrm{diag}(\omega)$ and $\Omega = (X^\top W X + \rho I)^{-1}$, $B_{ab} = -\Omega_{ab}/\Omega_{bb}$ for $a \neq b$ and $B_{bb} = 0$. A SLIM-type model [13] estimates a sparse non-negative matrix with zero diagonal,

$$\widehat{B} = \arg \min_{B \geq 0,\ \mathrm{diag}\, B = 0} \frac{1}{2N} \|X - XB\|_F^2 + \nu \Big[\xi \|B\|_1 + \frac{1-\xi}{2} \|B\|_F^2 \Big]. \tag{16}$$

For both, the score is the observed indicator vector multiplied by B, with a small prevalence tie-breaker. Personalised PageRank (PPR) [24] diffuses from the observed materials over the row-normalised direct graph Q: $\boldsymbol{z}^{(t+1)} = (1-\delta)\boldsymbol{u}_i + \delta\, \boldsymbol{z}^{(t)} Q$ with restart vector $\boldsymbol{u}_i$ supported on O_i, iterated 50 times. Formula k-nearest neighbours [25] and PureSVD [14] are included in the secondary protocol.

2.7. Formula Size and Composition

All ranking models share one size model and one composition rule, so that differences in F1 reflect ranking alone.

Size.

Let n_k be the (weighted) number of training formulas with k materials and $K_{\max}$ the largest training size. The size prior and its posterior given $|O_i| = o$ and known f are

$$p_0(k) = \frac{n_k + \frac{1}{2}}{N + K_{\max}/2}, \quad k = 1, \ldots, K_{\max}; \qquad p(k \mid o, f) \propto p_0(k)\, 1\{o(k, f) = o\}. \tag{17}$$

The indicator encodes the masking rule (2), which already implies $k \geq o + 1$. The prediction $\widehat{K}_i$ is the posterior median, and the top $\widehat{K}_i - o$ candidates form $\widehat{H}_i$. When f is withheld, the decoder must infer K_i from o and the size prior alone, which is harder. Central 80% posterior intervals $[L_i, U_i]$ are recalibrated with split conformal prediction [26,27] using the non-conformity score $e_i = \max\{L_i - K_i,\ K_i - U_i,\ 0\}$ on development-work-held-out instances; with n calibration instances the interval is widened by the empirical quantile of e at level $\lceil (n+1)(0.8) \rceil / n$.

Composition.

Observed fractions are kept, and the residual mass $m_i = 1 - \sum_{a \in O_i} w_{ia}$ is divided among the predicted hidden materials in proportion to a non-negative ratio score R_{ib} derived from the ten training formulas most similar to the observed part:

$$\widehat{w}_{ib} = \begin{cases} w_{ib}, & b \in O_i, \\ m_i R_{ib} \big/ \sum_{h \in \widehat{H}_i} R_{ih}, & b \in \widehat{H}_i, \\ 0, & \text{otherwise.} \end{cases} \tag{18}$$

2.8. Separating Source Variation from Estimation Noise

A poorly supported association can vary across books simply because each book provides few observations. We therefore examine a posterior-moment extension of SSCL that separates this estimation noise from between-book variation. This extension retains the same graph, source splits and ranking task. Its changes concern the quantities pooled across sources.

Write $A_{abd} = c_{abd} + \alpha\pi_{bd}$ and $B_{abd} = c_{ad} - c_{abd} + \alpha(1 - \pi_{bd})$. The beta distribution in Equation (8) gives exact conditional moments on the log-odds scale:

$$m_{abd} = \psi(A_{abd}) - \psi(B_{abd}) - \operatorname{logit}\pi_{bd}, \tag{19}$$

$$V_{abd} = \psi_1(A_{abd}) + \psi_1(B_{abd}), \tag{20}$$

where ψ and ψ_1 are the digamma and trigamma functions. These expressions follow by differentiating the beta log-normalising constant: $\mathrm{E}\log\theta = \psi(A) - \psi(A+B)$ and $\mathrm{E}\log(1-\theta) = \psi(B) - \psi(A+B)$. Taking their difference gives Equation (19); the corresponding covariance terms cancel to give Equation (20). The moments condition on the empirical-Bayes prevalence estimates, rather than integrating their estimation uncertainty.

For an edge with positive exposure, let $q_d = v_{ad}/\sum_e v_{ae}$ and $\bar{m}_{ab} = \sum_d q_d m_{abd}$. Suppose $m_{abd} = \mu_{ab} + u_d + e_d$, with independent, zero-mean source effects and errors, $\mathrm{Var}(u_d) = \tau^2_{ab}$ and $\mathrm{Var}(e_d) = V_{abd}$, treating the weights as fixed. Expanding the weighted sum of squares yields

$$\mathrm{E}\left[\sum_d q_d(m_{abd} - \bar{m}_{ab})^2\right] = \left(1 - \sum_d q_d^2\right)\tau^2_{ab} + \sum_d q_d(1-q_d)V_{abd}. \tag{21}$$

The first term measures genuine source variation; the second is the contribution of estimation noise. Solving this identity and truncating at zero gives the moment estimator

$$\widehat{\tau}^2_{ab} = \max\left\{0, \frac{\sum_d q_d(m_{abd} - \bar{m}_{ab})^2 - \sum_d q_d(1-q_d)V_{abd}}{1 - \sum_d q_d^2}\right\}. \tag{22}$$

For one exposed source we set $\widehat{\tau}^2 = 0$ and retain its posterior estimation variance. A new-source predictive scale is

$$U^2_{ab} = \widehat{\tau}^2_{ab} + \sum_d q_d^2(V_{abd} + \widehat{\tau}^2_{ab}), \qquad \widetilde{G}_{ab} = \operatorname{expit}(\operatorname{logit}\pi_b + \bar{m}_{ab} - \gamma U_{ab}). \tag{23}$$

The added term accounts for estimating the pooled mean. We blend $\widetilde{G}$ with the direct graph as in Equation (11). Nested validation selects among $\alpha \in \{0.5, 1, 2, 5\}$, $\lambda \in \{0, 0.5, 1\}$, $\gamma \in \{0, 0.5\}$ and mean, maximum or top-two reduction. The resulting 72 configurations include the direct graph as $\lambda = 0$. These are exploratory extensions made after the original benchmark was analysed; their results are reported separately in Section 3.7.

2.9. Completing a Residual Formula without Knowing Its Missing Fraction

A researcher normally knows the surviving materials and their relative amounts, but may know neither the original ingredient count nor the surviving fraction of the full mass. The deployment decoder therefore accepts relative observed amounts and estimates the two missing totals from training data. Let K be the full ingredient count, $m = |O|$, and

$p_{\text{train}}(K)$ the smoothed training histogram. Under a simple exchangeable retention model, each ingredient survives with probability ρ, and $\rho \sim \text{Beta}(a_0, b_0)$. Integrating over ρ gives

$$p(K \mid m) \;\propto\; p_{\text{train}}(K) \binom{K}{m} \frac{B(m + a_0, K - m + b_0)}{B(a_0, b_0)} \, 1\{K > m\}. \tag{24}$$

With $a_0 = b_0 = 1$, the likelihood reduces to $1/(K+1)$. We use its posterior median to minimise posterior expected absolute count error. The assumption describes unknown independent retention; the abundance and prefix tests examine different mechanisms. An investigator can replace the inferred count with a documented total.

For composition, write the supplied relative amounts as $o_a > 0$ and $u_a = o_a / \sum_{j \in O} o_j$. The same training-neighbour estimator used by the benchmark supplies a positive reference vector r on the selected support. If the known mass M is undocumented, estimate it as $\widehat{M} = \sum_{a \in O} r_a / \sum_{j \in \widehat{S}} r_j$. We then solve

$$\min_{x \in \Delta^{|\widehat{S}|-1}} \sum_{j \in \widehat{S}} x_j \log(x_j / r_j) \quad \text{subject to} \quad x_a = M u_a \quad (a \in O). \tag{25}$$

The normalisation of r changes the objective only by a constant. Strict convexity gives a unique solution for positive r and $0 < M < 1$. A Lagrange multiplier for the remaining mass gives $\log(x_b / r_b) + 1 + \nu = 0$ for each hidden material, hence

$$x_a = M u_a \quad (a \in O), \qquad x_b = (1 - M) \frac{r_b}{\sum_{j \in \widehat{S} \setminus O} r_j} \quad (b \notin O). \tag{26}$$

This makes the composition rule a constrained information projection: known ratios remain fixed while the residual mass follows the closest reference composition. The benchmark decoder is the special case in which the observed absolute mass is supplied.

To distinguish finding the right hidden mixture from guessing its total mass, we additionally report hidden-shape total variation and absolute missing-mass error. With p_H and $\widehat{p}_H$ normalised separately over all unobserved materials,

$$\text{TV}_H = \tfrac{1}{2} \sum_{b \notin O} |p_{Hb} - \widehat{p}_{Hb}|, \qquad E_M = \left| \sum_{b \notin O} p_b - \sum_{b \notin O} \widehat{p}_b \right|. \tag{27}$$

The operational test uses the original outer folds and masks, renormalises the visible amounts to sum to one before prediction, and withholds both the observation fraction and true visible mass. Its graph parameters are selected by the original inner benchmark. No query-specific hidden quantity enters inference.

2.10. Probability Calibration

Graph scores are not probabilities. For each outer fold we fit, on inner held-out-work predictions only, a temperature map and a two-parameter logistic (Platt-type) map [28,29],

$$p^{(t)} = \text{expit}\big(\text{logit}(p)/t\big), \qquad p^{(L)} = \text{expit}\big(a \, \text{logit}(p) + c\big), \quad a > 0, \tag{28}$$

by minimising candidate-level negative log-likelihood. The temperature map rescales log-odds about zero; the logistic map can also shift them. The outer test work contributes no calibration labels.

2.11. Evaluation Protocol

Primary protocol: nested leave-one-work-out.

Each of the six work groups serves once as the outer test work (Figure 3b). The test work is absent from vocabulary construction, prevalence and edge estimation, hyper-parameter selection and calibration. Inside the remaining five works, leave-one-work-out validation selects the configuration with the highest work-averaged hidden F1 (weighted Jaccard breaks ties), which is then refitted on all five works and applied to the test work. Nesting prevents selection bias in the reported error [30]; blocking by work follows the recommendation for grouped data [31] and the logic of domain-shift benchmarks [32,33]. The grids are: direct graph $\alpha \in \{0.5, 1, 2, 5\}$; SSCL $\alpha \in \{1, 2, 5\}$, $\gamma \in \{0, 0.25, 0.5, 1\}$, $\lambda \in \{0.25, 0.5, 0.75, 1\}$ and mean or maximum reduction; EASE ρ from 1 to 1000; SLIM four values of ν and three of ξ; PPR five values of δ.

Weak-supervision experiment.

The full nested protocol is repeated for the directed graph and SSCL with η added to the inner search. The admissible pool (14) is rebuilt for every inner and outer target, giving 62 to 142 admissible weak records per outer fold. Query eligibility and observed masks are generated once from the gold training vocabulary at each split and reused across every weak weight and model configuration, including inner selection. Additional weak vocabulary can expand the prediction space but cannot change the test input.

Secondary protocol: two works held out.

During development, five of the six work groups (85 formulas) were used; configurations were selected by leave-one-work-out validation and evaluated on all ten pairs of held-out works. This harder protocol removes two books at once and is used for the comparator panel, the near-duplicate test, the rarity analysis and the size intervals.

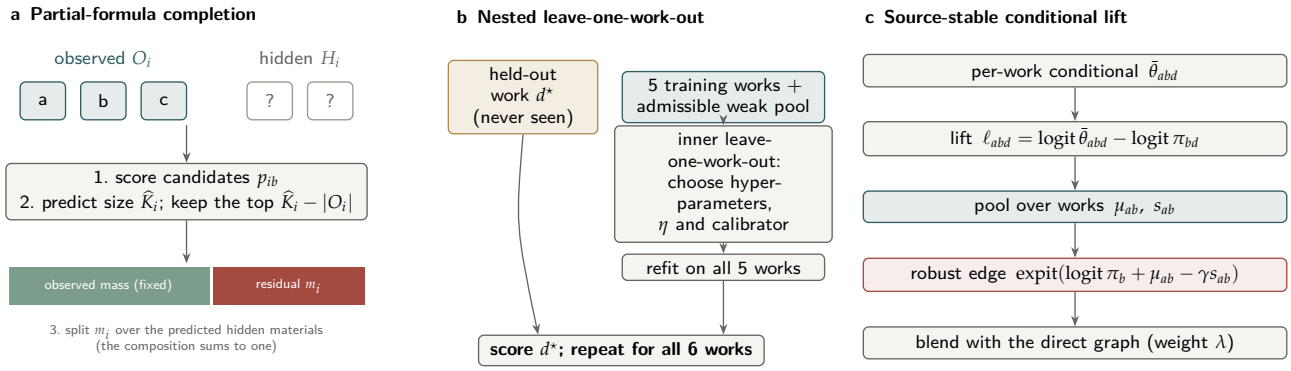


Figure 3. Task, protocol and method. (**a**) The model sees some materials with their reference mass fractions, scores all other materials, predicts the formula size, and divides the residual mass among the selected materials. (**b**) Nested leave-one-work-out evaluation: hyper-parameters, the weak-label weight η and the probability calibrator are chosen by inner validation on the five training works; the held-out work is scored once. (**c**) Source-stable conditional lift (Equations (7)–(11)).

2.12. Metrics

Set recovery.

With $\widehat{H}_i$ the predicted and H_i the true hidden set,

$$P_i = \frac{|H_i \cap \widehat{H}_i|}{|\widehat{H}_i|}, \qquad R_i = \frac{|H_i \cap \widehat{H}_i|}{|H_i|}, \qquad F_{1,i} = \frac{2P_iR_i}{P_i + R_i}, \tag{29}$$

with $F_{1,i} = 0$ when $P_i = R_i = 0$. Size error is $|\widehat{K}_i - K_i|$.

Composition.

The weighted Jaccard similarity [34] of two compositions is

$$J_i^w = \frac{\sum_j \min(w_{ij}, \widehat{w}_{ij})}{\sum_j \max(w_{ij}, \widehat{w}_{ij})}. \tag{30}$$

Lemma 1. *For $\boldsymbol{w}, \widehat{\boldsymbol{w}} \in \Delta^{M-1}$ with total-variation distance* $\mathrm{TV} = \frac{1}{2}\sum_j |w_j - \widehat{w}_j|$,

$$J^w = \frac{1 - \mathrm{TV}}{1 + \mathrm{TV}}. \tag{31}$$

If $\widehat{\boldsymbol{w}}$ is produced by rule (18), then $\mathrm{TV} \le m_i$ and hence $J_i^w \ge (1 - m_i)/(1 + m_i)$, whatever materials are selected.

Proof. Since $\min(x, y) + \max(x, y) = x + y$ and $\max(x, y) - \min(x, y) = |x - y|$, summing over j gives $\sum\min + \sum\max = 2$ and $\sum\max - \sum\min = 2\,\mathrm{TV}$, so $\sum\min = 1 - \mathrm{TV}$ and $\sum\max = 1 + \mathrm{TV}$. Under rule (18) the observed coordinates agree and both vectors place mass m_i on the remaining coordinates, so $\sum_{j \notin O_i} |w_{ij} - \widehat{w}_{ij}| \le 2m_i$. □

Lemma 1 shows that weighted Jaccard is a monotone transform of total variation, and that part of it is guaranteed by the observed input: with half of a typical formula's mass observed ($m_i = 0.5$) the score cannot fall below $1/3$. Weighted Jaccard should therefore be compared across models on the same masks, never against zero, and the quantity $1 - \mathrm{TV}_i / m_i \in [0, 1]$ isolates what the model earns on the hidden mass.

Calibration.

For candidate labels $y_n \in \{0, 1\}$ and probabilities p_n, $n = 1, \ldots, L$, we report the Brier score [35], the negative log-likelihood and the adaptive calibration error [36],

$$\mathrm{Brier} = \frac{1}{L}\sum_n (p_n - y_n)^2, \qquad \mathrm{NLL} = -\frac{1}{L}\sum_n \big[y_n \log p_n + (1 - y_n)\log(1 - p_n)\big],$$
$$\mathrm{ACE} = \sum_g \frac{|I_g|}{L}\Big|\frac{1}{|I_g|}\sum_{n \in I_g}(p_n - y_n)\Big|, \tag{32}$$

where the I_g are equal-count probability groups within each work. Brier score and NLL are averaged within formula, then within work; all three are then averaged over works.

2.13. Statistical Inference with Six Works

Formulas from the same work are dependent, and masks of the same formula are repeated measurements, so neither is an independent replicate. We therefore treat the six work-level differences Δ_{md} as the data. Under joint invariance of the work-level difference vector to independent sign changes, the two-sided sign-flip (randomisation) test [37] has

$$p_{\mathrm{flip}} = 2^{-D} \sum_{\boldsymbol{s} \in \{-1,1\}^D} 1\Big\{\Big|\frac{1}{D}\sum_d s_d \Delta_{md}\Big| \ge |\widehat{\Delta}_m|\Big\}. \tag{33}$$

Few clusters limit the resolution of this test [38], and the next statement quantifies the limit.

Proposition 3 (Attainable p-value floor)**.** *If $D' \ge 1$ of the D differences are non-zero, then $p_{\mathrm{flip}} \ge 2^{1-D'}$. If all differences are zero, $p_{\mathrm{flip}} = 1$. With $D = 6$ and one tie, $p_{\mathrm{flip}} \ge 0.0625$ however large the effect.*

Proof. The identity assignment and the global sign flip both give statistics equal to $|\widehat{\Delta}_m|$. Flipping the sign of a zero difference does not change the statistic, so each of these two patterns is repeated $2^{D-D'}$ times. Hence at least $2 \cdot 2^{D-D'}$ of the 2^D assignments meet the condition in Equation (33). □

For this comparison, the attainable two-sided sign-flip resolution is 0.0625. Alongside p_{flip}, we report the 90% confidence interval obtained by inverting the test,

$$\mathcal{C}_{0.90} = \{\theta : \; p_{\text{flip}}(\Delta_{m1} - \theta, \ldots, \Delta_{mD} - \theta) > 0.10\}, \tag{34}$$

under the same joint sign-invariance assumption about θ, together with the formula-weighted estimate and the range of leave-one-work-out means.

SSCL versus frequency on hidden F1 is the principal comparison in the reported benchmark. The directed graph against frequency, SSCL against the directed graph, and weak-augmented against gold-only SSCL are secondary comparisons. All other analyses are descriptive.

2.14. Diagnostics

Near duplicates.

A formula is a cross-work near duplicate if $\max_{r:\, d_r \neq d_i} |S_i \cap S_r| / |S_i \cup S_r| \geq 0.8$. The secondary protocol is repeated after removing such formulas.

Rarity and stability.

Hidden-material recall is stratified by the number of training formulas containing the material (absent, one, two to four, at least five). Stability is the mean pairwise Jaccard similarity of the five predicted hidden sets of a formula.

Case study.

One formula is displayed in full. From the six Hong formulas under mask seed 17, we select the formula whose hidden F1 is closest to the median, breaking ties by identifier.

The chemical-similarity analysis and the computational profile are described in Appendix A; notation is summarised in Table A2.

3. Results

3.1. Conditional Co-Formulation Transfers Across Books

Table 3 summarises the primary protocol. All structured models exceed frequency on precision, recall, hidden F1 and weighted Jaccard. SSCL reached a work-averaged hidden F1 of 0.426 against 0.378 for frequency, and the directed graph reached the same value. EASE attained the highest weighted Jaccard (0.473), closely followed by SSCL (0.472). SLIM gained least over frequency. Size error is identical across models by construction.

Table 3. Primary protocol: nested leave-one-work-out completion at 50% uniform observation. Entries are unweighted means of six work-level means (90 evaluated formulas, five masks each). All models share the size module, hence the common size error.

Model	Precision	Recall	Hidden F1	Weighted Jaccard	Size MAE
Frequency	0.432	0.363	0.378	0.451	1.145
Directed graph	0.481	0.413	0.426	0.470	1.145
SSCL	0.480	0.414	**0.426**	0.472	1.145
EASE	0.479	0.408	0.423	**0.473**	1.145
SLIM	0.440	0.371	0.387	0.455	1.145
PPR	0.467	0.397	0.412	0.459	1.145

The gain is consistent across books (Table 4, Figure 4). SSCL and the directed graph improved on frequency in five works and tied in Xiaolu, which contributes only two formulas. The largest gains occurred in the Hong group (SSCL +0.116), and Huizhai remained the hardest group for every model (F1 below 0.29).

Table 4. Hidden F1 by held-out work in the primary protocol. n is the number of evaluated formulas; each formula score averages five masks.

Held-out work	n	Frequency	Directed graph	SSCL	EASE
Chen	38	0.3812	0.4268	0.4199	0.4135
Hong	6	0.4689	0.5583	0.5850	0.5717
Huizhai	9	0.2268	0.2877	0.2877	0.2867
Liexiang	6	0.4365	0.4649	0.4490	0.4588
Xiaolu	2	0.4000	0.4000	0.4000	0.3833
Xiangcheng	29	0.3557	0.4180	0.4159	0.4232

Table 5 gives the work-level inference. For the primary comparison the mean difference is 0.048 with exact $p_{\text{flip}} = 0.0625$, which by Proposition 3 is the smallest value the design could produce after the Xiaolu tie. The 90% sign-flip interval, 0.013 to 0.088, excludes zero. The formula-weighted estimate (0.050) and the leave-one-work-out means (0.035 to 0.058) show that the result depends neither on the weighting nor on any single book. SSCL and the directed graph differ by 0.0003 ($p = 1.0$), and the 90% interval (-0.009 to 0.013) bounds the effect of the dispersion penalty to about one hundredth of F1 in either direction. Five of the six outer folds selected a positive penalty γ, so the null result reflects an inert penalty rather than a penalty that was never chosen.

Table 5. Work-level inference for hidden F1 (primary protocol, $D = 6$ works). The column $+/0/-$ counts works that improve, tie and worsen. p is the exact two-sided sign-flip p-value (Equation (33)); the interval inverts the same test (Equation (34)). LOWO: range of the mean after leaving out one work.

Comparison	Mean $\widehat{\Delta}$	Formula-weighted	$+/0/-$	p	90% interval	LOWO range
SSCL − frequency (primary)	0.048	0.050	5/1/0	0.0625	[0.013, 0.088]	[0.035, 0.058]
Directed graph − frequency	0.048	0.053	5/1/0	0.0625	[0.023, 0.075]	[0.039, 0.057]
EASE − frequency	0.045	0.049	5/0/1	0.0625	[0.008, 0.081]	[0.033, 0.057]
SSCL − directed graph	0.000	−0.003	1/2/3	1.000	[−0.009, 0.013]	—
SSCL weak − SSCL gold	−0.020	—	2/2/2	0.500	[-0.051, 0.005]	—

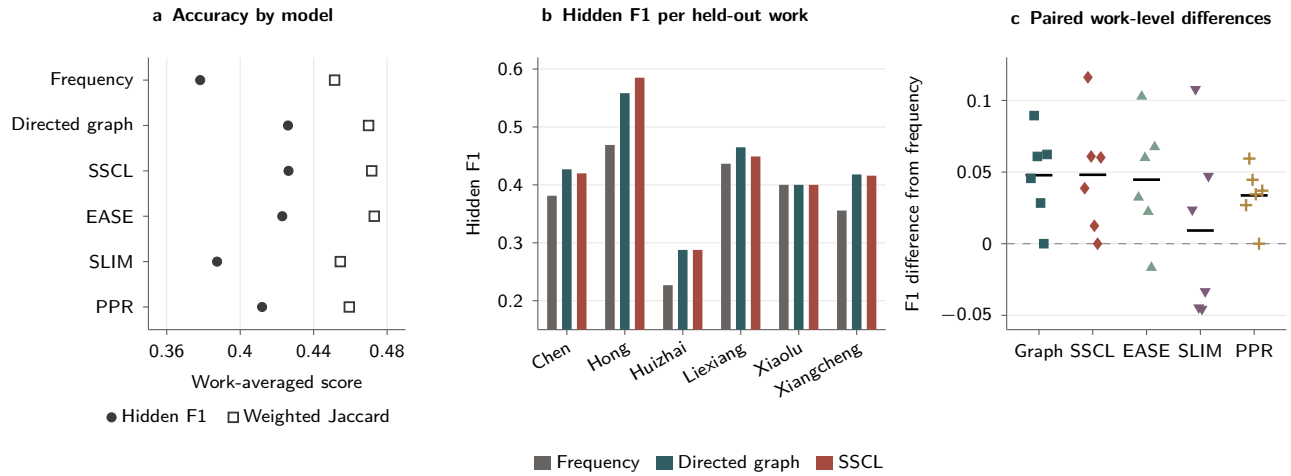


Figure 4. Primary protocol. (**a**) Work-averaged hidden F1 (filled circles) and weighted Jaccard (open squares) for six models. (**b**) Hidden F1 per held-out work for frequency, the directed graph and SSCL. (**c**) Paired work-level differences from frequency; points are held-out works and bars are means. For the directed graph and SSCL five works improve and one ties (exact sign-flip $p = 0.0625$, the attainable minimum).

3.2. Weak Records under Matched Observation Masks

Depending on which collection was blocked, an outer fold admitted 62 to 142 weak records (Table 6). Inner validation selected a positive SSCL weight in five of six folds. Effective weak mass ranged from zero in Liexiang to 98.67 formula units in Hong. The family denominator is fixed before collection blocking: it counts all eligible, ratio-ready weak witnesses in the family after gold families have been removed. A family with blocked witnesses therefore contributes only its admissible fraction of the budget.

Table 6. Weak-data exposure selected for SSCL in each outer fold. Effective mass is Equation (15) at the selected η.

Held-out work	Gold training formulas	Admissible weak records	Selected η	Effective weak mass
Chen	53	88	0.50	31.58
Hong	85	142	1.00	98.67
Huizhai	81	62	1.00	38.17
Liexiang	85	62	0.00	0.00
Xiaolu	89	62	0.10	3.82
Xiangcheng	62	62	1.00	38.17

Under identical masks, weak-record augmentation changes directed-graph F1 from 0.4260 to 0.4258 and SSCL F1 from 0.4263 to 0.4065 (Table 7). For SSCL, the largest decrease occurs in Huizhai (-0.1013), followed by Hong (-0.0267); Chen and Xiangcheng improve by about 0.005, while Liexiang and Xiaolu tie. The work-mean difference is -0.0197 with sign-flip $p = 0.5$. Both systems retain count MAE 1.1447. Their selected weak weights differ, but the size posterior produces the same count decisions on these queries. A larger weighted sample can alter a histogram without changing its posterior median.

Table 7. Weak supervision in the primary protocol. Gold-only and gold-plus-weak systems are evaluated on the same 90 gold formulas with identical observed ingredients; η is selected inside each outer fold.

Training evidence	Precision	Recall	Hidden F1	Weighted Jaccard	Size MAE
Directed graph, gold	0.481	0.413	0.426	0.470	1.145
Directed graph, gold + weak	0.477	0.415	0.426	0.466	1.145
SSCL, gold	0.480	0.414	0.426	0.472	1.145
SSCL, gold + weak	0.465	0.388	0.407	0.449	1.145

3.3. Robustness: Two Books Held Out and Near Duplicates Removed

Holding out two books at once lowers every score but preserves the pattern (Table 8, Figure 5a). The directed graph reached hidden F1 0.367 against 0.339 for frequency (a gain of 0.029) and had the highest weighted Jaccard (0.449); SSCL (0.365), EASE (0.363) and PPR (0.362) followed closely. Formula *k*-nearest neighbours (0.343) and PureSVD (0.334) were close to frequency, which indicates that the useful signal lies in item-to-item dependencies rather than in whole-formula similarity or a low-rank structure.

Table 8. Secondary protocol: two held-out works (all ten pairs of five development works: Chen, 38 formulas; Huizhai, 10; Liexiang, 6; Xiaolu, 2; and unresolved Xiangcheng, 29) at 50% uniform observation. Entries are work-averaged means.

Model	Precision	Recall	Hidden F1	Weighted Jaccard
Frequency	0.405	0.316	0.339	0.440
Formula *k*NN	0.408	0.323	0.343	0.444
PureSVD	0.395	0.316	0.334	0.420
Directed graph	0.437	0.344	**0.367**	**0.449**
SSCL	0.430	0.345	0.365	0.442
EASE	0.432	0.340	0.363	0.445
SLIM	0.383	0.312	0.327	0.441
PPR	0.432	0.338	0.362	0.445

Nine development formulas have a near duplicate (material-set Jaccard at least 0.8) in another work. The near-duplicate test was run with the development-stage configurations of the directed graph and of the probabilistic decoder described in Appendix A. Removing the nine formulas lowered all scores by about 0.017 but left the gain of the graph over frequency essentially unchanged: 0.027 before and 0.026 after removal (frequency 0.339 $\rightarrow$ 0.322, graph 0.3654 $\rightarrow$ 0.3482, decoder 0.3665 $\rightarrow$ 0.3494; Figure 5b). The transferable signal is therefore not an artefact of copied recipes that escaped genealogy filtering. Predicted hidden sets were moderately stable across masks (mean pairwise Jaccard 0.644): which half of a formula is visible changes the completion, but most selected materials persist.

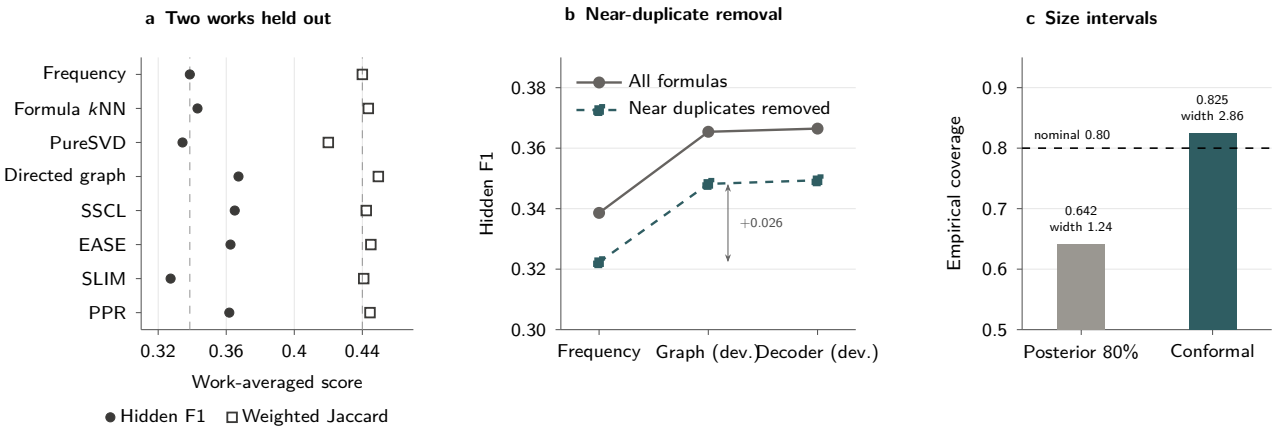


Figure 5. Secondary protocol (two held-out works). (**a**) Hidden F1 and weighted Jaccard for eight models (Table 8); dashed lines mark frequency. (**b**) Hidden F1 before and after removing nine formulas with a cross-work near duplicate, for frequency and the development-stage graph and decoder; the gain of the graph over frequency is preserved. (**c**) Empirical coverage and mean width of central 80% size intervals before and after split-conformal expansion; the dashed line marks the nominal level.

3.4. The Observation Mechanism Changes Difficulty More Than Ranking

At 50% observation, every structured model beat frequency under all four mechanisms (Table 9). The differences among the structured models were small (at most 0.017), and the best model changed with the mechanism: the directed graph under abundance-biased retention, EASE under source-prefix retention, and SSCL under uniform retention and when f was withheld. Given the resolution of six works, these differences in ordering should not be over-interpreted; the robust conclusion is that conditioning on observed materials helps under every mechanism tested.

Table 9. Work-averaged hidden F1 under four observation mechanisms at 50% observation.

Mechanism	Frequency	Directed graph	SSCL	EASE
Uniform	0.378	0.426	**0.426**	0.423
Abundance-biased	0.389	**0.430**	0.426	0.429
Source-list prefix	0.399	0.468	0.466	**0.481**
Uniform, f withheld	0.378	0.392	**0.404**	0.387

The observed fraction changes the task itself (Figure 6). Under uniform, abundance-biased and f-withheld retention, hidden F1 falls as more of the formula is observed, because the hidden set shrinks to one or two materials that are disproportionately rare, and a single miss then costs a large share of F1. At 75% uniform retention all models fell to about 0.27–0.29, and at 75% abundance-biased retention frequency (0.355) matched or exceeded the structured models: once the principal ingredients are visible, what remains is a rare accent that no co-occurrence model predicts. Source-prefix retention behaves differently. Its curves are flat for SSCL (0.462 at 75% versus 0.395 for frequency), consistent with base materials tending to be listed first, so that a prefix reveals the base from which the remaining materials can be inferred.

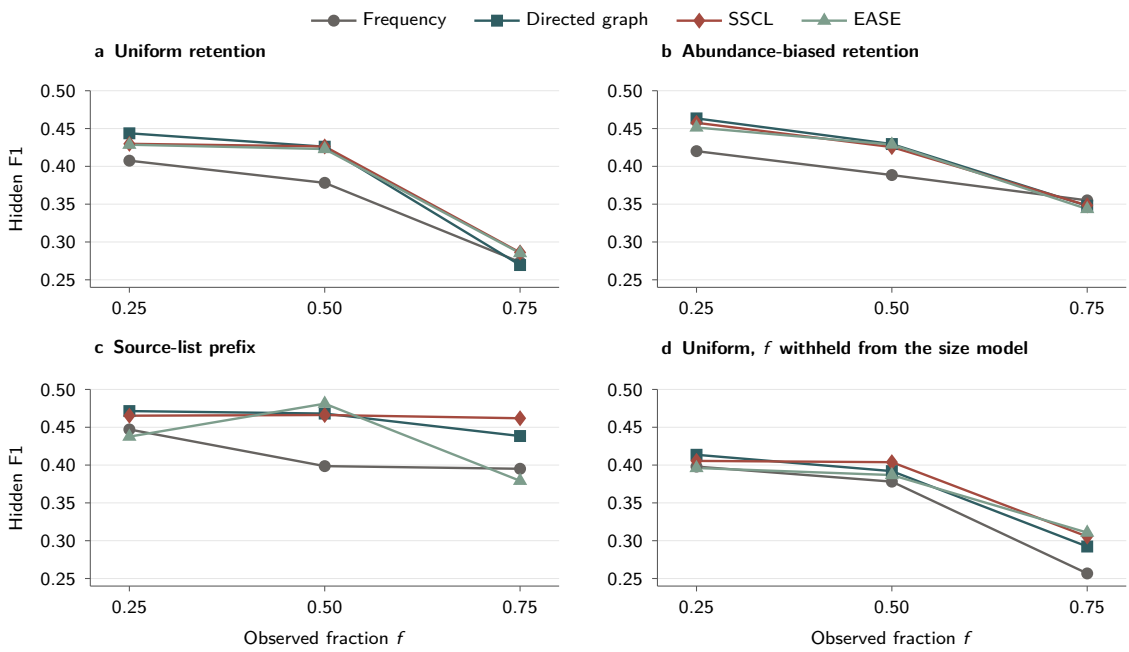


Figure 6. Hidden F1 as a function of the observed fraction under four observation mechanisms. (**a**) Uniform retention. (**b**) Retention probability proportional to reference mass. (**c**) The first materials in source order. (**d**) Uniform retention with f withheld from the size model. Points are work-averaged means over the six outer folds of the primary protocol; shared axes show changes in both difficulty and ordering.

3.5. Calibrated Probabilities and Size Intervals

Raw SSCL scores were overconfident: the 186 candidates scored above 0.8 had a mean score of 0.839 but were present only 60.8% of the time (Table 10, Figure 7). Temperature scaling lowered every error metric but sharpened the scores, so that 455 candidates exceeded 0.8 with an empirical frequency of only 0.527. This is the expected failure of a map that can only rescale log-odds about zero. The two-parameter logistic map, which can also shift them, reduced the Brier score from 0.0427 to 0.0340, NLL from 0.1572 to 0.1262 and ACE from 0.0579 to 0.0145, and left only two scores above 0.8. For users, a calibrated score has a concrete meaning: among candidates scored p in books not used for fitting, a fraction of about p belongs to the formula.

Table 10. Nested calibration of SSCL candidate scores (work-averaged). The last three columns describe predictions with $p > 0.8$.

Method	Brier	NLL	ACE	Count	Mean p	Observed rate
Uncalibrated	0.0427	0.1572	0.0579	186	0.839	0.608
Temperature	0.0406	0.1456	0.0371	455	0.874	0.527
Logistic	**0.0340**	**0.1262**	**0.0145**	2	0.831	1.000

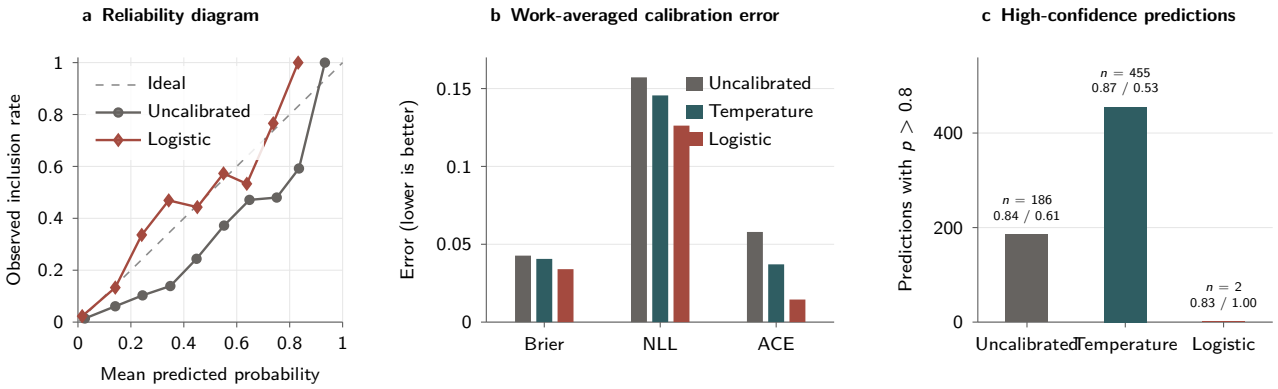


Figure 7. Nested probability calibration of SSCL scores. (**a**) Pooled reliability diagram before and after logistic calibration. (**b**) Work-averaged Brier score, negative log-likelihood and adaptive calibration error. (**c**) Number of predictions above 0.8, labelled with mean probability/empirical frequency. The two logistic-calibrated predictions above 0.8 are too few for a stable estimate.

For formula size, the central 80% posterior interval covered the true size in only 64.2% of 1660 instances of the secondary protocol (mean width 1.24 materials), because the posterior ignores between-book variation in formula length. Split-conformal expansion, calibrated on 420 development instances ($\lceil 421 \times 0.8 \rceil / 420 = 0.8024$), widened every interval by one material and raised coverage to 82.5% at a mean width of 2.86 (Figure 5c). The finite-sample guarantee of conformal prediction assumes exchangeability of calibration and test instances, which work-blocked data satisfy only approximately; the empirical coverage is nevertheless close to nominal.

3.6. Where the Errors Come From

Rarity.

The secondary protocol contains 6450 hidden-material opportunities across 1660 evaluated fold–mask–formula instances (Table 11). Their recall decomposes as a weighted average over frequency strata, $R = \sum_k \omega_k R_k$, where ω_k is the share of opportunities in stratum k. Materials seen in at least five training formulas account for 60.1% of opportunities and 97.7% of all recovered materials (recall 0.573). Recall collapses to 0.056 for materials seen two to four times and 0.023 for materials seen once. Materials absent from the training vocabulary make up 22.4% of opportunities and can never be predicted, so the recall of a closed-vocabulary ranking model is bounded by $1 - 0.224 = 0.776$ in this protocol. Rarity explains more error than formula size: formulas with at least eight materials reached hidden F1 0.446, compared with 0.352 for two to four materials and 0.344 for five to seven.

Table 11. Hidden-material recall by training frequency (secondary protocol, pooled over folds and masks; used for error localisation, not inference).

Training frequency	Opportunities	Share	Recovered	Recall
Absent from vocabulary	1445	0.224	0	0.000
One formula	342	0.053	8	0.023
Two to four formulas	787	0.122	44	0.056
At least five formulas	3876	0.601	2221	0.573
All	6450	1.000	2273	0.352

False positives concentrate on the hubs of the co-formulation graph: white sandalwood (462 false additions), borneol (409), musk (378) and generic agarwood (376). Frequent false negatives include scrophularia root, spikenard, Zhan-grade agarwood, frank-

incense and clove. The model learns productive hubs and over-uses them when a target book follows a different convention.

A worked example.

The selected case is the Hong *yinxiang* formula introduced in Section 1 (Table 12). With six of eleven materials observed, the model predicted the correct size and recovered three of five hidden materials (Zhan-grade agarwood, borneol and white sandalwood), missed frankincense and licorice, and added clove and *linglingxiang*; precision, recall and F1 all equal 0.60. Proportions were less accurate than identities: Zhan-grade agarwood received 0.117 of the mass against 0.332 in the source, and borneol 0.099 against 0.013. The weighted Jaccard is 0.483; by Lemma 1 the observed 45.3% of the mass alone guarantees at least $0.453/1.547 = 0.293$. The hidden-mass score is $1 - \mathrm{TV}/m_i = 0.36$: of the 54.7% of the mass that was hidden, the prediction placed about a third on the correct materials.

Table 12. Completion of the Hong *yinxiang* formula (mask seed 17). Observed materials keep their reference fractions; a dash marks a material that is absent from the reference or from the prediction. Reference fractions are rounded and sum to 0.996.

Material	Outcome	Reference	Predicted
Agarwood	observed	0.066	0.066
Aromatic grass	observed	0.266	0.266
Musk	observed	0.053	0.053
Onycha	observed	0.001	0.001
Patchouli	observed	0.001	0.001
Spikenard	observed	0.066	0.066
Zhan-grade agarwood	recovered	0.332	0.117
Borneol	recovered	0.013	0.099
White sandalwood	recovered	0.066	0.103
Frankincense	missed	0.066	—
Licorice	missed	0.066	—
Clove	falsely added	—	0.106
Linglingxiang	falsely added	—	0.120

Composition rule.

Lemma 1 also explains a development result. Replacing an unconstrained composition decoder by rule (18), which keeps the observed fractions and closes the prediction to the residual mass, raised weighted Jaccard in the secondary protocol from 0.387 to 0.427 without changing a single selected material. Part of this gain is the guaranteed floor that the rule enforces, which is why we recommend reporting $1 - \mathrm{TV}_i/m_i$ alongside weighted Jaccard.

Chemistry.

A chemical-similarity kernel built from specimen-level analytical profiles of 18 materials neither helped nor hurt completion (hidden F1 0.3905 with the kernel, 0.3911 without; Appendix A). The 18 profiled materials represent only 22.2% of material types but 65.2% of aromatic mentions in the ratio-ready corpus, so the null result plausibly concerns common materials whose co-formulation is already well estimated from the texts.

3.7. Posterior Moments and Operational Reconstruction

Table 13 separates the posterior-moment extension from the operational decoder. The former reaches hidden F1 0.4266, compared with 0.4263 for SSCL and 0.4260 for the directed graph. Inner validation selects direct-graph fallback in four outer folds and posterior-moment pooling in two. The similar aggregate performance shows that exact posterior moments sharpen the model's interpretation without establishing a substantial

accuracy gain on this corpus. We retain SSCL for deployment and release the posterior extension as a tested alternative.

With both the observation fraction and visible mass withheld, deployed SSCL reaches hidden F1 0.3943 and weighted Jaccard 0.3845. Its missing-mass error is 0.2560 and its hidden-shape total variation is 0.7209. These two measures identify different challenges: total-mass estimation loses information supplied by the benchmark, while rare missing materials remain difficult to identify. The interface consequently allows a researcher to supply a documented ingredient count or visible mass when either is available, and preserves the supplied within-fragment ratios exactly.

Table 13. V5 extensions on the same six outer work folds (90 formulas, five masks each). Lower is better for count MAE, hidden-shape TV and missing-mass error. The first three rows receive the benchmark observation fraction and visible absolute mass; the operational row receives neither.

Model	Hidden F1	W. Jaccard	Count MAE	TV_H	E_M
Directed graph	0.4260	0.4699	1.1447	0.7125	0.0000
SSCL	0.4263	0.4716	1.1447	0.7075	0.0000
Posterior-moment extension	0.4266	0.4708	1.1447	0.7111	0.0000
SSCL, unknown fraction and mass	0.3943	0.3845	1.7318	0.7209	0.2560

The released web implementation uses a frozen graph fitted to the 91 reviewed records, with parameters selected by leave-one-work-out validation over these training records. Its predictions retain model version, observed materials, supplied constraints and supporting formula references. Saved records remain separate from the benchmark and are never incorporated into training automatically. This keeps new scholarly corrections traceable while the released evaluation remains reproducible.

4. Discussion

4.1. What Transfers Between Books

Across six held-out books, knowing part of a formula improves the prediction of the rest by about 0.05 in F1, roughly a 13% relative improvement over frequency, and the improvement appeared in every book where the methods differed. It survives holding out two books at once, removal of near duplicates and four observation mechanisms. The directed graph, SSCL, EASE and PPR all capture it, whereas formula-level nearest neighbours and low-rank factorisation largely do not. The transferable object is therefore the pairwise conditional structure of co-formulation: which materials habitually accompany which, independently of the compiler. It is consistent with the common description of incense formulas as a base of woods and resins completed by characteristic accents, and it gives heritage scholars a quantitative way to flag likely lacunae in damaged or abridged witnesses.

4.2. Why the Heterogeneity Penalty Is Inert

SSCL was designed to prefer associations that hold across books. It matched but did not beat the directed graph, and the 90% interval limits the penalty's effect to about ± 0.01 F1. Three observations explain this. First, by Remark 1, s_{ab} is estimated from at most five training works (four in inner folds), so its relative error is of order 35–40%, and the ranking perturbation it induces is largely noise. Second, the edges that matter for recall are those between common materials (Table 11), and these are estimated precisely and consistently in every book, leaving little heterogeneity to penalise. Third, the edges that are genuinely book-specific involve rare materials that neither model recovers. Proposition 1 suggests when SSCL should help: with more source domains, s_{ab} becomes informative, and the

lower-quantile interpretation offers a principled choice of γ through the desired predictive level. The penalty is therefore a hypothesis to retest as the corpus grows rather than a component to discard.

4.3. Scale and Quality Are Separate Variables

The matched-mask experiment makes the role of source review concrete. Adding 146 unreviewed records supplies more observed pairs, yet leaves size decisions unchanged and lowers SSCL recovery, principally in Huizhai. Positive inner-selected weights describe what validation selected within the available training works; their outer-fold performance measures whether that choice transfers. Here it does not improve aggregate recovery. Review should prioritise families containing rare materials, families spanning collections and additional witnesses of unstable edges, because these records can add information that the present graph lacks.

4.4. Lessons for Evaluating Historical Reconstruction

Three design choices changed conclusions and should be standard. Blocking by work and by collection measures transfer to a new book rather than memorisation of a known one. Reporting the observation mechanism is necessary because the observed fraction and mechanism change task difficulty more than they change model ranking. Composition scores need a reference point: by Lemma 1, weighted Jaccard contains a floor determined by the observed mass, and the hidden-mass score $1 - \mathrm{TV}/m$ removes it. Finally, the number of books, not the number of formulas, governs statistical precision. With six works and one tie, this two-sided sign-flip test cannot reach $p < 0.0625$ (Proposition 3); reporting randomisation intervals and leave-one-work-out ranges conveys the evidence more faithfully than a binary significance verdict.

4.5. Limitations

The gold set was reviewed by a single curator, so inter-annotator agreement on formula boundaries, material identities and quantities is not yet available. The six work groups are unequal in size, and two contain two and six formulas; conclusions about individual books, particularly Xiaolu, are correspondingly weak. Observation mechanisms were simulated; empirical damage patterns from parallel witnesses would provide more realistic masks. The near-duplicate analysis, the composition-rule comparison and the chemistry ablation used development-stage configurations (Appendix A) rather than the final tuned models, and are reported as supporting evidence only. Finally, canonical material identities for historical names are themselves uncertain for part of the vocabulary; the recall ceiling in Table 11 partly reflects this uncertainty.

4.6. Future Work

Four extensions follow directly. (i) Reviewing the 146 weak records and adding further public-domain incense collections would increase both the gold set and, more importantly, the number of work domains, which is what limits inference and the usefulness of the SSCL penalty. (ii) A second independent annotation of a stratified sample would quantify label reliability and support a probabilistic label model for the weak pool [17]. (iii) Parallel witnesses can be aligned to learn an empirical deletion model $p(O_i \mid S_i, d_i)$ to replace simulated masks. (iv) Proportions should be modelled on the simplex, for example by logistic-normal or Dirichlet regression [20] with the co-formulation graph as a structured prior, and chemical information should enter through directed interaction features relevant to combustion rather than symmetric similarity.

5. Conclusions

Restoring fragmentary historical incense formulas can be posed as weighted set completion under source shift and evaluated honestly by holding out whole books. On a provenance-indexed corpus with 91 reviewed formulas from six work groups, conditional co-formulation structure transfers between books and improves hidden-ingredient F1 from 0.378 to 0.426 over global frequency, consistently across works. A source-stable variant with a robust-optimisation interpretation matches but does not improve on the plain directed graph at six domains. Unreviewed-record augmentation does not improve the matched-mask benchmark, nested calibration makes graph scores interpretable as probabilities, and material rarity sets the main remaining limit. The benchmark, its evidence hierarchy and its inference conventions are intended as a reusable basis for quantitative reconstruction of historical recipes.

Author Contributions: Conceptualization, H.Z.; methodology, H.Z.; software, H.Z.; validation, H.Z.; formal analysis, H.Z.; investigation, H.Z.; data curation, H.Z.; writing—original draft preparation, H.Z.; writing—review and editing, H.Z.; visualization, H.Z. The author has read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable. The study analyses public historical texts and published chemical records.

Informed Consent Statement: Not applicable.

Data Availability Statement: The accompanying reproducibility archive contains curated formula records, source locators, deterministic splits and masks, model implementations, evaluation outputs and figure source data. Records retain the source attribution and rights statement of each transcription. The operational interface is available at https://incense-trace-lab.joe-zhuhb.chatgpt.site/.

Acknowledgments: Generative AI tools, including OpenAI GPT-5.6 Sol and GPT-6 Astra and Anthropic Claude, assisted with historical-source retrieval, manuscript organisation and editing, mathematical development and checks, software implementation, figure preparation, repository organisation and typesetting. Human authorship and responsibility for the research remain with Hongbai Zhu.

Conflicts of Interest: The author declares no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

ACE	Adaptive calibration error
EASE	Embarrassingly shallow autoencoder
LOWO	Leave one work out
MAE	Mean absolute error
NLL	Negative log-likelihood
PPR	Personalised PageRank
SLIM	Sparse linear method
SSCL	Source-stable conditional lift
TV	Total variation

Appendix A. Development-Stage Analyses

Probabilistic decoder.

Before the final graph models were tuned, a probabilistic decoder was developed that scores each candidate with a logistic model on graph-derived features and averages

predictions over 200 posterior parameter draws. It is not part of the main comparison. It served as the host model for the chemistry ablation, appears as "decoder (dev.)" in the near-duplicate test (Figure 5b), and was used for the composition-rule comparison in Section 3.6.

Chemical similarity.

A chemical registry links 18 of the 81 canonical materials to specimen-level analytical profiles drawn from 17 publications and 40 PubChem compounds, covering, among others, agarwood, sandalwood, frankincense, clove, cassia and cyperus [39–44]. Profiles record taxon, plant part, preparation, analytical method and measurement basis; peak-area data, calibrated concentrations and presence-only reports are kept as separate evidence types. For compound sets $\mathcal{P}_a, \mathcal{P}_b$ with Morgan fingerprint bit sets B_u [45], the fingerprint similarity is $T_{ab} = \max_{u\in\mathcal{P}_a, v\in\mathcal{P}_b} |B_u \cap B_v|/|B_u \cup B_v|$. Physicochemical vectors (molecular weight, hydrogen-bond donors and acceptors, rotatable bonds, topological polar surface area, log$P$) and controlled odour-tag vectors (woody, balsamic, spicy, floral, camphoraceous, animalic and others) are abundance-weighted where abundances are available, and compared by cosine similarity. The kernel

$$\mathcal{K}_{ab} = \omega_f T_{ab} + \omega_x \max\left(0, C_{ab}^{\text{phys}}\right) + \omega_o \max\left(0, C_{ab}^{\text{odor}}\right), \qquad \omega_f + \omega_x + \omega_o = 1, \tag{A1}$$

was added to the decoder for the 18 covered materials. Table A1 reports the ablation and an inverse-odour task in which 44 formulas are reconstructed from their odour-tag profile.

Table A1. Chemistry ablations (development stage). Completion reports hidden F1; inverse odour reports weighted Jaccard on 44 formulas.

Task	Model	Score
Completion	Decoder without chemistry	0.3911
Completion	Decoder with chemical kernel	0.3905
Inverse odour	Odour-profile kNN	0.2913
Inverse odour	Forward-consistent decoder	0.2824
Inverse odour	Decoder without chemistry	0.2835
Inverse odour	Shuffled chemical profiles	0.2863

Computational profile.

Dense graph construction costs $O(NM^2)$, or $O(N\bar{K}^2)$ when pairs are accumulated sparsely; scoring one query costs $O(|O_i|M)$ plus $O(M)$ expected time for top-K selection. The implementation uses NumPy and scikit-learn [46] with dense arrays ($M \leq 81$). On one Apple arm64 process with data loaded, the development decoder, which is more expensive than the graph models, fitted 85 formulas and 80 materials in a median 0.115 s (interquartile range 0.005 s, 11 runs), and one prediction with 200 posterior draws took a median 1.70 ms (interquartile range 0.14 ms). The graph models avoid optimisation and sampling and are faster still.

Appendix B. Notation

Table A2. Principal notation.

Symbol	Meaning
$i; a, b; d$	formula; observed and candidate materials; work group
$S_i, O_i, H_i; K_i$	complete, observed and hidden material sets; formula size
$q_{ij}, w_{ij}; m_i$	converted mass and closed fraction; residual (hidden) mass
$f; o(K, f)$	requested observed fraction; number of observed materials
π_b, π_{bd}	global and work-specific smoothed prevalence
G_{ab}	directed conditional edge
$\bar{\theta}_{abd}, \ell_{abd}, v_{ad}$	within-work conditional, lift and exposure weight
$\mu_{ab}, s_{ab}; G^{\star}_{ab}, G^{\mathrm{SS}}_{ab}$	pooled lift and dispersion; robust and blended SSCL edges
$\alpha, \kappa, \gamma, \lambda$	smoothing, work shrinkage, dispersion penalty, blend weight
$z_i, \varphi(i), C(i); \omega_i, \eta$	evidence class, family, collection; record weight, weak-label weight
$\mathcal{W}_{d^\star}$	admissible weak pool for test work $d^\star$
$p_{ib}; \widehat{K}_i, \widehat{H}_i, \widehat{\boldsymbol{w}}_i$	candidate score; predicted size, hidden set and composition
$\Delta_{md}; D$	work-level difference from frequency; number of works
$A_{abd}, B_{abd}; \psi, \psi_1$	beta shape parameters; digamma and trigamma functions
$m_{abd}, V_{abd}; q_d$	posterior lift mean and variance; normalised source weight
$\widehat{\tau}^2_{ab}, U^2_{ab}$	estimated source variance; predictive-scale variance
$\rho; a_0, b_0$	retention probability; beta retention-prior parameters
M, u_a, r_j	visible mass; observed relative share; reference composition weight
TV_H, E_M	hidden-shape total variation; missing-mass absolute error

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